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Editor-in-Chief
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Dear Editor,

We submit for consideration the manuscript *"MMSE Pair Scheduling: Sharp Complexity, Certified Power Allocation, and Regime-Dependent Rate Gains"* by Pooya Hejazi and Masoud Sabaei.

The paper studies joint user pairing and power allocation in an uplink with linear MMSE reception and no successive interference cancellation. It makes four contributions. First, it proves fixed-power scheduling hardness with three receive dimensions and a fixed rational channel alphabet. Second, it derives the sharp whole-quadrant concavity threshold for the two-user rate and characterizes rectangular power pricing by at most seventeen candidates. Third, it develops a finite-box/integral certificate that closes to 0.01 nat on 27 of 32 stored ray-traced cases, with a 97.82% median reduction in the fixed-anchor gap. Fourth, it shows why earlier pairing benchmarks were flat: at $M = 8$, MMSE reception removes the two-user interference and the fixed-power pairing gain is zero in all 32 cases. On a standardized 3GPP TR 38.901 CDL-A/B/C benchmark at $K = 16$ and $M = 2$, the proposed exact assignment and power search improves the strongest common-model baseline by a median 1.31%, or 0.129 bit/s/Hz, with 53 wins in 53 cases and a 95% interval $[1.19, 1.42]$%.

The paper is explicit about its boundary. The CDL rate gains are feasible and verified but not globally certified; the certified result remains the $M = 8$ bound. The exact search is compared with full SLSQP, adapted I-WMMSE and matching–WMMSE under the same power box, budget and service constraints. A 30-second baseline sensitivity leaves the proposed result ahead in all twelve tested cases. The manuscript is accompanied by a supplementary document and a reproducible package containing the CDL generator, case files, result tables, plotting scripts and the finite-box verification code.

This work has not been published elsewhere and is not under consideration by another journal. Both authors have approved the manuscript and declare no conflict of interest. AI tools were used for code review and manuscript text revision; the authors are responsible for the content.

Sincerely,

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